

Shiying Xiao

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Education

M.Sc. in Statistics

University of Connecticut
Advisor: [Prof. Jun Yan](#)

August 2023 – May 2026
Storrs, CT

M.Econ. in Statistics

Shanxi University of Finance and Economics
Advisor: [Prof. Tao Wang](#)

September 2019 – July 2022
Taiyuan, China

B.Econ. in Insurance

Shanxi University of Finance and Economics

September 2015 – July 2019
Taiyuan, China

Research Interests

Applied econometrics · Biostatistics · High-dimensional data · Input-output analysis · National economic accounting
· Network analysis · Omics data analysis · Public health applications · Statistical computing

Publications (* refers to the corresponding author)

- Thornton, S., Szymanski, M. R., Brewer, G. J., Casa, D. J., Moore, T. E., [Xiao, S.](#), Green, A. A., Pardee, K. and Lee, E. C.* (2026). Sample processing methods affect salivary metabolomics in human exercise-stress studies. *Metabolomics*, 22(4): 123. [DOI](#)
- Syednejad, A., Sacko, T. J., Babigian, C. J., Moore, T. E., [Xiao, S.](#), Liddle, J. C. and Sartor, G. C.* (2026). Cocaine-induced changes in the BRD4 interactome identifies casein kinase 1 epsilon as a therapeutic target. *Journal of Neurochemistry*, 170(1): e70336. [DOI](#)
- Wang, T.* and [Xiao, S.](#) (2022). A comparative study of domestic industrial circulation between China and the United States from the perspective of input-output network. *Statistical Research*, 39(11): 32-43. [URL](#) (in Chinese with English abstract)
- [Xiao, S.*](#), Yan, J. and Zhang, P. (2022). Incorporating auxiliary information in betweenness measure for input-output networks. *Physica A: Statistical Mechanics and its Applications*, 607: 128200. [DOI](#)
- Wang, T., [Xiao, S.](#), Yan, J. and Zhang, P.* (2021). Regional and sectoral structures of the Chinese economy: A network perspective from multi-regional input-output tables. *Physica A: Statistical Mechanics and its Applications*, 581: 126196. [DOI](#)

Manuscripts

- Aladelokun, O., Benitez, K., Shen, X., Jain, A., Berardi, D., Lu, L., Gardner, E., Yang, J., Almazwini, M., Rattray, N. J. W., Butensky, S., [Xiao, S.](#), Siddique, S., Berner, A. M., Aalizadeh, R., Vasiliou, V., Paty, P. B., Garcia-Milian, R., Khan, S. A. and Johnson, C. H.* (2026+). Sex-specific metabolic phenotypes in early onset colorectal cancer. Under review.
- Chen, C.*, [Xiao, S.](#) and Salike, N. (2026+). China's export supply chain resilience measures and improvement strategy: The application of product space theory. (in Chinese with English abstract)
- Wang, T., [Xiao, S.*](#) and Yan, J. (2026+). Comparison of sectoral structures between China and Japan: A network perspective. [arXiv](#). Under review.
- [Xiao, S.*](#), Yan, J. and Zhang, P. (2026+). Groupwise regularized adaptive sparse precision solution.

5. **Xiao, S.**, Liddle, J., Balsbaugh, J., Watt, C. and Moore, T.* (2026+). msDiaLogue: Analyzing and visualizing DIA proteomics data in R.
6. Zhang, P.*, **Xiao, S.**, Robb, W. H., Liu, D., Jefferson, A. L. and Yan, J. (2026+). Gaussian graphical models for functional connectivity analysis: A statistical review with applications to Alzheimer’s disease. [arXiv](#). Under review.

Presentations and Posters

1. Functional Connectivity Estimation Based on Gaussian Graphical Models: A Statistical Review and Applications to Alzheimer’s Disease Data. Datablitz speaker. The 27th Annual Neuroscience at Storrs Symposium. University of Connecticut, Storrs, CT, 10/22/2024.
2. Gaussian Graphical Models for Functional Connectivity Analysis: A Statistical Review and Applications to Alzheimer’s Disease Data. Poster presenter. The 33rd Annual Applied Statistics Symposium for the International Chinese Statistical Association (ICSA 2024). Nashville, TN, 06/17/2024.
3. Functional Connectivity Analysis in Brain Networks: A Statistical Review and Application to Alzheimer’s Disease Data. Invited speaker. The 37th New England Statistics Symposium (NESS 2024). University of Connecticut, Storrs, CT, 05/23/2024.
4. Introduction to Data Analysis in R for DIA-based Proteomics Using msDiaLogue. Assistant instructor. The 4th Annual Introduction to Mass Spectrometry-Based Proteomics Workshop. University of Connecticut, Storrs, CT, 03/15/2024.

Software

1. **Xiao, S.**, Yan, J. and Zhang, P. (2026). **grasps**: Groupwise Regularized Adaptive Sparse Precision Solution. R package version 0.1.1, <https://CRAN.R-project.org/package=grasps>.
2. **Xiao, S.**, Yan, J. and Zhang, P. (2024). **ionet**: Network Analysis for Input-Output Tables. R package version 0.2.2, <https://CRAN.R-project.org/package=ionet>.
3. **Xiao, S.**, Moore, T. and Watt, C. (2025). **msDiaLogue**: Analysis and Visuals for Data-Independent Acquisition Mass Spectrometry Data. R package version 0.0.7, <https://uconn-scs.github.io/msDiaLogue>.
4. **Xiao, S.** (2025). **pecan**: Portfolio for Economic Complexity Analysis and Navigation. R package version 0.1.0, <https://CRAN.R-project.org/package=pecan>.
5. **Xiao, S.**, Yan, J. and Zhang, P. (2026). **spice**: Sparse Precision (Inverse Covariance) Estimation. R package version 0.1.0, <https://shiyang-xiao.com/spice>.

Experience

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|--|-------------------------|
| Research Technician | July 2026 – Present |
| Department of Educational Leadership, University of Connecticut
Storrs, CT | |
| <ul style="list-style-type: none"> Conducted empirical analyses of Connecticut’s magnet school lottery system and educational outcomes. | |
| Statistical Consultant | August 2023 – May 2026 |
| Statistical Consulting Services, University of Connecticut
Storrs, CT | |
| <ul style="list-style-type: none"> Provided statistical consulting and analytical support to researchers across various disciplines. | |
| Research Assistant | August 2023 – May 2026 |
| Department of Statistics, University of Connecticut
Storrs, CT | |
| <ul style="list-style-type: none"> Conducted research on high-dimensional statistical methods and developed R packages. | |
| Assistant Professor of Teaching | August 2022 – July 2023 |
| Fuzhou University of International Studies and Trade
Fuzhou, China | |
| <ul style="list-style-type: none"> Designed and taught undergraduate courses in statistics, data visualization, and business analytics. | |

Research Assistant

Shanxi University of Finance and Economics

June 2019 – July 2022

Taiyuan, China

- Contributed to grant-funded research in economic statistics, network analysis, and policy evaluation.

Reviewer Service

- Journal of Open Source Software
- Physica A: Statistical Mechanics and its Applications

Technical Skills

- **Programming:** R, Python, C++
- **Computing:** Linux, Bash, High-Performance Computing (HPC)
- **Tools:** Git, L^AT_EX, Markdown, Quarto